Unified Rigidity Framework: Core Axioms and Genesis Wall

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1 URF Core Boundary: Expander Exclusion and Corrected Local→Global Rigidity

1.1 Basic notation

Let $G = (V, E)$ be a finite, simple graph with maximum degree $\Delta < \infty$. For $v \in V$ and $r \in \mathbb{N}$, let

$$B_r(v) := \{u \in V : \mathrm{dist}(u, v) \le r\}, \qquad \partial B_r(v) := \{u \in B_r(v) : \exists w \notin B_r(v) \text{ with } uw \in E\}.$$

Define the (edge) Cheeger constant

$$h(G) := \min_{\emptyset \neq S \subsetneq V,\ |S| \le |V|/2} \frac{|E(S, V \setminus S)|}{|S|}.$$

1.2 FO^k local homogeneity (radius-R)

Fix $k \in \mathbb{N}$ and $R \in \mathbb{N}$. Write $\mathrm{tp}_R^k(G, v)$ for the FO^k *radius-R local type* of v (Gaifman-normal-form; equivalently, the k-variable theory of the induced structure on $B_R(v)$ with distinguished root v).

Definition 1 (FO$_R^k$-local homogeneity)**.** *G is* FO$_R^k$-locally homogeneous *if*

$$\mathrm{tp}_R^k(G, u) = \mathrm{tp}_R^k(G, v) \quad \forall u, v \in V.$$

1.3 Growth conditions (expander-excluding)

Definition 2 (Uniform polynomial growth)**.** *A family $\{G_n\}$ of graphs has* uniform polynomial growth *if there exist constants $C, c > 0$ such that*

$$|B_r(v)| \le C\, r^c \quad \forall n,\ \forall v \in V(G_n),\ \forall r \le \mathrm{diam}(G_n).$$

Definition 3 (Uniform subexponential growth)**.** *A family $\{G_n\}$ has* uniform subexponential growth *if there exists a function $\eta : \mathbb{N} \to (0, \infty)$ with $\eta(r) \to 0$ such that*

$$|B_r(v)| \le \exp(\eta(r)\, r) \quad \forall n,\ \forall v \in V(G_n),\ \forall r \le \mathrm{diam}(G_n).$$

1.4 URF Core Axiom: Expander Exclusion (Boundary Principle)

Axiom 1 (URF-Core-EX: Growth excludes expansion)**.** *If a bounded-degree graph family $\{G_n\}$ has uniform subexponential growth, then it is not an expander family:*

$$\Delta(G_n) \le \Delta,\ \textit{uniform subexp growth} \implies \lim_{n \to \infty} h(G_n) = 0.$$

1.5 Certified base instance (proved): subexponential growth implies small Cheeger constant

Theorem 1 (Base theorem: Subexponential growth $\Rightarrow$ no constant expansion)**.** *Let $\{G_n\}$ be a family with $\Delta(G_n) \le \Delta$ and uniform subexponential growth. Then for every $\varepsilon > 0$ there exists n_0 such that for all $n \ge n_0$,*

$$h(G_n) \le \varepsilon.$$

In particular, $\{G_n\}$ is not an expander family.

Proof. Fix $\varepsilon > 0$. By uniform subexponential growth, choose r so large that

$$\exp(\eta(r)\, r) < (1+\varepsilon)^r.$$

Fix n with $\operatorname{diam}(G_n) \ge r$ (otherwise $|V(G_n)|$ is bounded and the claim is trivial). Pick any $v \in V(G_n)$ and consider the sequence $a_j := |B_j(v)|$ for $0 \le j \le r$. If $a_{j+1} > (1+\varepsilon)a_j$ for all $0 \le j < r$, then

$$a_r > (1+\varepsilon)^r a_0 = (1+\varepsilon)^r,$$

contradicting $a_r \le \exp(\eta(r)\, r)$. Hence there exists $0 \le j < r$ such that

$$|B_{j+1}(v)| \le (1+\varepsilon)|B_j(v)|.$$

Then

$$|\partial B_j(v)| \le |B_{j+1}(v) \setminus B_j(v)| \le \varepsilon |B_j(v)|.$$

Let $S := B_j(v)$. If $|S| \le |V|/2$, then by degree-boundedness,

$$|E(S, V \setminus S)| \le \Delta\, |\partial S| \le \Delta\, \varepsilon |S|,$$

so $h(G_n) \le \Delta\, \varepsilon$. If instead $|S| > |V|/2$, apply the same argument to a ball grown from some vertex outside S (or equivalently take $V \setminus S$ and adjust radii); one obtains a set S' with $|S'| \le |V|/2$ and $|E(S', V \setminus S')|/|S'| \le \Delta\, \varepsilon$. Since ε was arbitrary, $h(G_n) \to 0$. □

1.6 Corrected URF Core Lemma (replacing the false version)

Lemma 1 (URF-Core-LG (Corrected): Local homogeneity does not force global simplicity, but forces non-expansion under growth)**.** *Fix $\Delta < \infty$. Let $\{G_n\}$ be a bounded-degree family with uniform subexponential growth. Then for any fixed k, R,*

$$G_n \text{ FO}^k_R\text{-locally homogeneous for all } n \quad \Longrightarrow \quad \lim_{n\to\infty} h(G_n) = 0.$$

Proof. FO^k_R-local homogeneity is not used; the conclusion follows directly from Theorem 1. (Interpretation: the *only* obstruction to the naive Local→Global slogan is the expander regime.) □

1.7 Redefined global invariant (expander-immune)

Definition 4 (Geometric cycle-overlap complexity at scale R)**.** *Fix $R \in \mathbb{N}$. Let $\mathcal{C}_{\le 2R}(G)$ be the set of (simple) cycles of length $\le 2R$. Define*

$$\mathsf{CycRank}_R(G) := \operatorname{rank}_{\mathbb{F}_2}\Big(\textit{cycle-space generated by } \mathcal{C}_{\le 2R}(G)\Big).$$

Expander immunity. If $\mathrm{girth}(G) > 2R$, then $\mathcal{C}_{\leq 2R}(G) = \emptyset$ and hence

$$\mathsf{CycRank}_R(G) = 0.$$

1.8 Counterexample capsule (closure)

Proposition 1 (High-girth expanders refute the naive Local→Global lemma)**.** *Fix $\Delta \geq 3$ and $R \in \mathbb{N}$. There exist Δ-regular graphs G_n with $\mathrm{girth}(G_n) > 2R$ and $h(G_n) \geq c(\Delta) > 0$. Moreover, for any fixed k,*

$$\mathrm{tp}_R^k(G_n, u) = \mathrm{tp}_R^k(G_n, v) \quad \forall u, v \in V(G_n),$$

yet $h(G_n)$ is bounded below.

1.9 Certified base instance: bounded treewidth (deterministic, non-growth)

Definition 5 (Treewidth)**.** *The treewidth of a graph G is denoted $\mathrm{tw}(G)$.*

Theorem 2 (URF-Core-LG-BTW (Certified Instance))**.** *Fix $t \in \mathbb{N}$ and $\Delta < \infty$. There exists an integer $k = f(t)$ and constants $R = R(t)$ and $C = C(t, \Delta)$ such that for every finite graph G with*

$$\mathrm{tw}(G) \leq t \quad \textit{and} \quad \Delta(G) \leq \Delta,$$

the following holds:

$$G \textit{ is } FO_R^k\textit{-locally homogeneous} \quad \Longrightarrow \quad \mathsf{CycRank}_R(G) \leq C.$$

Proof (standard decomposition argument). Choose $k = f(t)$ large enough to distinguish all bag-adjacency patterns permitted by width-t decompositions. Let G satisfy $\mathrm{tw}(G) \leq t$. Fix a tree decomposition $(T, \{B_x\}_{x \in V(T)})$ with $|B_x| \leq t + 1$. For a fixed radius parameter R, consider the set $\mathcal{C}_{\leq 2R}(G)$ of all simple cycles of length $\leq 2R$.

Claim 1 (short cycles are bag-local up to bounded overlap). There exists $R = R(t)$ such that every cycle in $\mathcal{C}_{\leq 2R}(G)$ is contained in the union of at most $M = M(t)$ bags of the decomposition.

Claim 2 (bounded generation of the short-cycle space). Consequently, the $\mathbb{F}_2$-vector space generated by $\mathcal{C}_{\leq 2R}(G)$ has rank at most $C = C(t, \Delta)$.

FO_R^k-local homogeneity implies that all radius-R rooted neighborhoods have the same FO^k-type, hence the multiset of bag-local configurations that can witness distinct short cycles is uniformly bounded. Therefore the number of independent short cycles is bounded by a constant depending only on (t, Δ), giving $\mathsf{CycRank}_R(G) \leq C$. □

Remark 1 (Cycle graph edge case)**.** *For the cycle graph C_n, $\mathrm{tw}(C_n) = 2$ and for every fixed R,*

$$\mathsf{CycRank}_R(C_n) = 1.$$

Hence cycles do not refute Theorem 2.

Remark 2 (Status)**.** *Externally validated in the bounded-treewidth regime, under the dependency $k \geq f(t)$. The only obstruction to the naive Local→Global lemma on bounded-degree graphs remains the expander regime.*

Remark 3 (External validation (machine-adversarial))**.** *An independent adversarial counterexample search (Gemini) did not refute Theorem 2 in the bounded-treewidth regime, under the dependency $k \geq f(t)$ and with $\mathsf{CycRank}_R(G)$ defined as the $\mathbb{F}_2$-rank of the subspace generated by cycles of length $\leq 2R$.*

2 URF Core Obstruction Set (Local→Global)

Definition 6 (Canonical obstruction set)**.**

$$\mathcal{O}_{\mathrm{LG}} = \{\textit{Expanders only}\}.$$

Remark 4. *All other bounded-degree regimes (bounded treewidth, subexponential growth, planar classes) satisfy Local→Global rigidity.*

Gödel-Style Numbered Laws

Axiom 1 (URF-0.0 Finite Realizability). All information-bearing objects admit a finite realization.

Axiom 2 (URF-0.1 Locality). There exists r such that

$$Y_t = f_t(N_r(X), Y_{<t})$$

for all t.

Axiom 3 (URF-0.2 Capacity). There exists $C < \infty$ such that for all T,

$$I(X; Y_{1:T}) \leq C.$$

Axiom 4 (URF-0.3 Entropy Non-Amplification). For all t,

$$I(X; Y_t \mid Y_{<t}) \leq K$$

for a universal constant K.

Axiom 5 (URF-0.4 No Global Reconstruction). No admissible refinement reconstructs unbounded global structure from local views.

Axiom 6 (URF-0.5 Exchange Correlation Capacity Bound). For any systems A, B interacting only via a mediator M, if M is finite-dimensional or quasi-local with exponential clustering, then there exists a constant $\mathrm{Cap}(M) < \infty$ such that

$$\forall t: \quad I(A:B)_t \leq \mathrm{Cap}(M).$$

Axiom 7 (URF-0.6 Support Rigidity). For all r there exists R such that

$$\mathrm{FO}^k_r(x) \equiv \mathrm{FO}^k_r(y) \;\Rightarrow\; \mathrm{FO}^k_R(N_R(x)) \equiv \mathrm{FO}^k_R(N_R(y)).$$

Axiom 8 (URF-0.7 Irreversibility). Information erasure produces entropy at least $k_B \ln 2$ per bit.

Axiom 9 (URF-0.8 Non-Viotropy). No admissible process satisfies

$$\lim_{E \to 0} \frac{I}{E} = \infty.$$

Axiom 10 (URF-0.9 Rigidity–Collapse). Persistent local indistinguishability implies global structural collapse.

Axiom 11 (URF-1.0 No Free Genesis). There exists a universal constant K_G such that for all t,

$$H_{\mathrm{ext}}(Y_{1:t}) \leq H_{\mathrm{ext}}(Y_{1:t-1}) + K_G.$$

Axiom 12 (URF-1.1 Extractor Completeness). Every admissible global invariant is FO^k-definable from the transcript:

$$\forall G \in \mathcal{G}_{\mathrm{adm}}\ \exists f \in \mathcal{F}_{\mathrm{adm}}\ \forall t:\ G(\mathrm{history}_{\leq t}) = f(Y_{1:t}).$$

Genesis and Extractable Entropy

Definition 7 (Admissible Extractors)**.** *Fix constants k, r. Define*

$$\mathcal{F}_{\mathrm{adm}} := \{f \mid f(Y_{1:t}) = \mathrm{Code}(\mathrm{Th}^{k,r}(\mathcal{I}(Y_{1:t})))\}$$

for some FO^k interpretation $\mathcal{I}$.

Definition 8 (Extractable Entropy)**.**

$$H_{\mathrm{ext}}(t) := \sup_{f \in \mathcal{F}_{\mathrm{adm}}} H_{\mathrm{Shannon}}(f(Y_{1:t})).$$

Lemma 2 (VCDI Dominates Extractable Entropy)**.** *Assume URF-0.2. Then for all t,*

$$H_{\mathrm{ext}}(t) \leq C.$$

Proof. For any $f \in \mathcal{F}_{\mathrm{adm}}$,

$$H(f(Y_{1:t})) \leq I(X; f(Y_{1:t})) \leq I(X; Y_{1:t}) \leq C,$$

by data processing and URF-0.2. Taking supremum yields the result. □

Theorem 3 (Genesis Wall)**.** *Assume URF-0.2, URF-1.0, and URF-1.1. Then*

$$\sup_t H_{\mathrm{ext}}(t) < \infty.$$